

Consistency by Construction: Cross-Encoders with Reversal-Aware Decoding for SemEval-2027 Task 2 (DiCo-NLI)

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Abstract—DiCo-NLI asks a system to label an ordered phrase pair as equivalence, forward entailment, backward entailment or another relation, and then checks whether the labels it gives to (a, b) and (b, a) agree. We fine-tune multilingual cross-encoders and move consistency out of training and into decoding: the log-probabilities of a pair and its reversed twin are mapped into one frame, averaged, and decoded once, so every pair predicted as reversible is consistent by construction. On the development set, a seven-model ensemble reaches a Weighted F1 of 80.8, SoftCons of 95.7 and HardCons of 88.0, averaged over the English, Spanish, Basque and mixed tracks. Almost all remaining errors separate the negative class from the three entailment labels. Five methods aimed at that decision, including focal loss, extra iSTS data and a stacked classifier, did not beat the plain ensemble. If the reversible/negative split is read from the layout of the input files, scores rise to 92.6, 100 and 91.2, above the current development leader. We report that variant separately because it depends on how the data were packaged.

Index Terms—natural language inference, directional consistency, cross-encoders, multilingual NLP, Basque, SemEval

I. INTRODUCTION

Natural language inference (NLI) models are usually scored one example at a time. A model can do well on that score and still contradict itself, for example by labelling both (a, b) and (b, a) as forward entailment, which would make each phrase strictly more specific than the other. SemEval-2027 Task 2, DiCo-NLI [1], [2], makes this visible. Every reversible phrase pair appears in both orders, and systems are scored on per-example Weighted F1 and on two consistency measures over the pairs.

Our system is built around one observation: consistency can be enforced at decoding time for free, because both directions of every reversible pair are in the same input file. We average the two directions’ scores in a shared frame and choose a single label for the pair. This leaves the classifier with one job, which is to separate the entailment family from everything else. That job turned out to be the hard part. The contributions of this report are the decoding scheme and an exactly reversal-equivariant variant of the cross-encoder, a comparison of four multilingual backbones, and a negative result: none of the five NEG-focused methods we tried improved on a plain ensemble. Code, configurations and run logs are public.¹

II. TASK AND DATA

Each instance is an ordered pair (a, b) of short phrases, about six words in total, taken from the chunk alignments of

PhrasIS [3], which in turn come from the interpretable STS datasets of SemEval 2015–2016 [4]–[6]. The label is EQ, FE ($a \models b$), BE ($b \models a$) or NEG, where NEG covers every other PhrasIS relation (similar, related, opposite). The reversal map is $\text{Rev}(\text{EQ}) = \text{EQ}$, $\text{Rev}(\text{FE}) = \text{BE}$ and $\text{Rev}(\text{BE}) = \text{FE}$; NEG has no reverse. There are four tracks: English (T1), Spanish (T2), Basque (T3) and mixed-language pairs (T4).

Weighted F1 is the support-weighted F1 over the four labels. SoftCons counts a gold-reversible pair as consistent when both predictions are reversible and Rev-compatible, whatever the gold label. HardCons requires both directions to be correct. Predicting NEG on either side of a gold-reversible pair therefore costs both consistency scores at once, which makes a false NEG the most expensive error.

We checked three facts on every training and development file. Instance ids have the form `<pair_id>__<l1>-<l2>__<original|flipped>`, and the reversed twin swaps both the language codes and the direction tag. Every reversible instance has its twin in the same file, while every NEG instance appears once. A `pair_id` is shared across tracks, with two rows in each monolingual track and twelve in T4. Each monolingual track has 3,042 training rows (1,282 reversible pairs and 478 NEG) and 660 development rows (277 and 106).

III. RELATED WORK

Interpretable STS [5], [6] introduced the chunk-level relation types from which PhrasIS and DiCo-NLI are derived. Consistency of neural models under logical transformations has been studied for NLI [7], [8], usually through extra loss terms. For symmetric pair tasks, Xu et al. [9] and Kumar and Joshi [10] penalise disagreement between the two input orders; we use the same idea for an antisymmetric label set, where the two orders should agree up to the Rev map. Our NEG study draws on work showing that NLI models lean on lexical overlap [11] and fail on simple lexical inferences [12], and on focal loss [13] and intermediate-task transfer [14] as remedies. For Basque we rely on multilingual encoders; XLM-R-large was the strongest of the multilingual encoders evaluated on XNLIeu [15].

IV. SYSTEM

A. Cross-encoder

A pretrained encoder reads `[CLS] a [SEP] b [SEP]` and a linear head maps the first-token state to four logits [16], [17].

¹<https://github.com/MinLee0210/dico-nli-challenge>

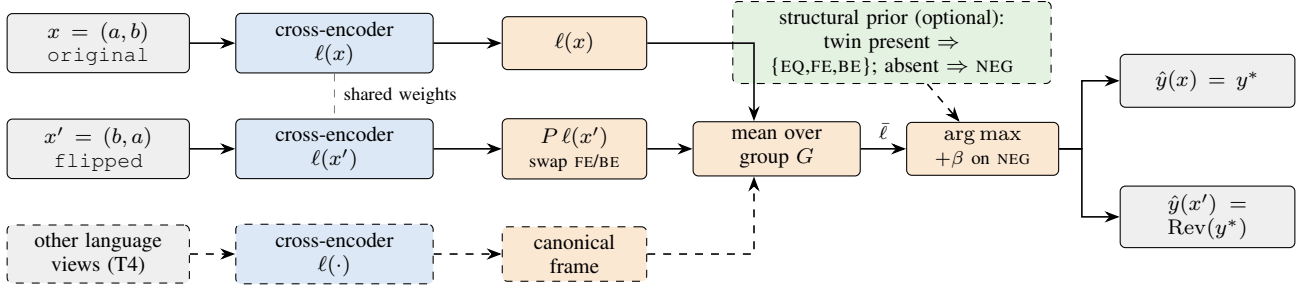


Fig. 1. Consistency-aware decoding. Scores of an instance and its reversed twin (and, in T4, of other language views of the same source pair) are mapped into the original frame, averaged, and decoded once. The reversed row receives the reversed label, so the pair is consistent whenever it is not NEG. The optional structural prior restricts the choice using the file layout.

We use DeBERTa-v3 [18], [19] for English only, and three multilingual backbones: mmBERT-base [20], an mDeBERTa-v3-base checkpoint fine-tuned on multilingual NLI data [21], and XLM-R-large [22] fine-tuned on XNLI [23], [24]. Multilingual models are trained on all four tracks at once (27,378 rows), so each source pair is seen in nine language combinations.

B. Consistency-aware decoding

Fig. 1 shows the decoding pipeline. For an instance x with log-probabilities $\ell(x) \in \mathbb{R}^4$, the canonical score is $\tilde{\ell}(x) = \ell(x)$ for original rows and $\tilde{\ell}(x) = P \ell(x)$ for flipped rows, where the permutation P swaps the FE and BE coordinates. Instances are grouped and the mean of $\tilde{\ell}$ over a group is decoded to one canonical label, after adding a bias β to the NEG score:

$$y^* = \arg \max_c \left(\frac{1}{|G|} \sum_{x \in G} \tilde{\ell}_c(x) + \beta [c = \text{NEG}] \right). \quad (1)$$

Original rows receive y^* and flipped rows receive $\text{Rev}(y^*)$. With *independent* grouping each instance is its own group, which is plain argmax. *Twin* grouping joins an instance with its reversed twin. *Source* grouping joins every row with the same `pair_id` in the decoded file, which pools language views in T4. Any group holding both directions is Rev-consistent by construction. All per-track results in this report decode each track file on its own.

C. Symmetric architecture

Decoding gives consistency only when both directions are in the input. We also built a model that is consistent for any single input. It scores both orders with the same encoder f and combines them as

$$g(a, b) = \frac{1}{2} (f(a, b) + P f(b, a)), \quad (2)$$

so $g(b, a) = P g(a, b)$ holds exactly (Fig. 2). The cost is two forward passes per example. A unit test checks the identity to within 10^{-5} .

D. Training objectives

The loss is cross-entropy, optionally with label smoothing, plus a reversal-consistency term between each row and its

twin, which a twin-aware batch sampler keeps in the same batch:

$$\mathcal{L} = \text{CE} + \lambda \text{KL}_{\text{sym}}(p(x) \| P p(x')). \quad (3)$$

For the NEG study we added class-weighted and focal [13] variants of the cross-entropy term and a *reverse-negative* augmentation that adds the reversed copy of each NEG pair; NEG is its own reverse, so the label stays correct.

E. Structural prior

The second data fact makes the reversible-or-NEG decision readable from the file: a group whose reversed twin is present is decoded among EQ, FE and BE, and a group without a twin is decoded as NEG. This is exact on training and development data. It is a property of how the files were built, not of the phrases, and the test files may be packaged differently, so we report every result with and without it. *Prior-aware* models are trained on the three reversible labels only, since under the prior the model is never asked about NEG; they must not be used without it.

V. EXPERIMENTAL SETUP

All models were trained on one NVIDIA RTX 4000 Ada (20 GB) with mixed precision, AdamW [25], 10% linear warm-up and linear decay, batch size 16–32 and maximum length 64–128 subword tokens. Learning rates were $2 \cdot 10^{-5}$ for DeBERTa-v3-base, $5 \cdot 10^{-5}$ for mmBERT, $1.5 \cdot 10^{-5}$ for mDeBERTa and 10^{-5} for XLM-R-large. The symmetric XLM-R model used batch size 8 to fit in memory. The multilingual models reach their best score within the first epoch, so for the main runs we validate every 15% of an epoch and stop after five checks without improvement.

The early-stopping monitor is the mean of the three official scores with the four development files decoded together. It is only used to stop training and to compare runs quickly; every table decodes each track separately. Ensembles average log-probabilities before decoding. Our metric code agrees with the official scorer to machine precision on all tracks and decoding modes, and the headline results were re-scored with the official scorer.

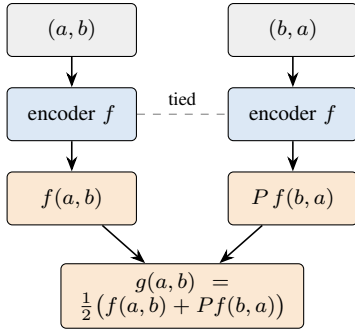


Fig. 2. The symmetric model scores both input orders with one encoder and averages the results after swapping FE and BE on the reversed side. Swapping a and b therefore swaps FE and BE in the output exactly.

TABLE I
DECODING MODES FOR ONE DEBERTA-v3-BASE MODEL ON THE T1 DEVELOPMENT SET. BEST VALUE PER COLUMN IN BOLD.

Decoding	W-F1	SoftCons	HardCons
independent	82.7	88.1	83.8
twin	82.1	93.5	86.3
source	82.1	93.5	86.3
twin + structural prior	92.4	100.0	91.0

TABLE II
METHODS AIMED AT THE NEGATIVE CLASS: ALL-4 DEVELOPMENT SCORES WITHOUT THE PRIOR. BEST VALUE PER COLUMN IN BOLD.

System	W-F1	SoftCons	HardCons
A (mDeBERTa)	79.4	95.9	87.0
+ NEG weight 2	79.7	91.2	83.1
+ focal, $\gamma = 2$	78.2	94.4	85.9
+ reverse negatives	80.2	91.2	83.1
+ iSTS answers-students	77.8	96.2	87.0
B (XLM-R-large)	78.8	96.3	86.3
+ reverse negatives	74.3	92.7	79.6
+ iSTS answers-students	77.6	92.8	84.3
Clean-7 ensemble	80.8	95.7	88.0
+ stacked NEG detector	81.4	91.8	84.7
with $\beta = -1.0$	79.1	98.5	89.8

VI. RESULTS

A. Decoding

Table I isolates decoding on a single DeBERTa-v3-base model for T1. Twin decoding costs 0.6 F1 points and gains 5.4 SoftCons and 2.5 HardCons points. SoftCons stays below 100 only because some reversible pairs are predicted as NEG. With the structural prior, the same model scores exactly what the development leader reports for T1 (92.4 / 100 / 91.0), which suggests the leader uses the same prior.

B. Backbones and model variants

Tables III and IV give per-track development scores without and with the structural prior. The two NLI-trained encoders, mDeBERTa and XLM-R-large, are close to each other and well ahead of mmBERT, which starts from masked-language-model weights and overfits after one epoch. A Basque-only

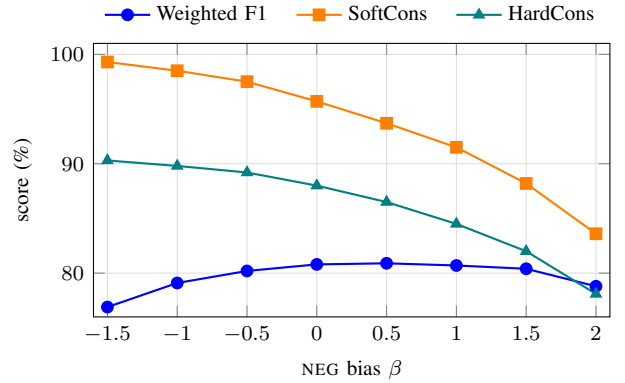


Fig. 3. Effect of the NEG bias β on the clean seven-model ensemble (ALL-4 development scores). Lowering β trades F1 for consistency.

encoder trained on T3 alone (JaunBERT) reached a monitor score of only 76.9, so we kept the multilingual models for Basque. The symmetric architecture does not help a single model, and adding it to the clean ensemble lowers HardCons from 88.0 to 87.3. With consistency already enforced at decoding time, the architectural guarantee adds nothing here.

On T1, the English-only DeBERTa-v3-large reaches 83.3 / 95.3 / 88.1, but the multilingual ensemble already scores 83.7 / 95.3 / 89.9 and adding the large model does not change it. Seed and backbone ensembles give the only consistent gains. The best clean system averages seven models (four mDeBERTa and three XLM-R-large seeds) and reaches 80.8 / 95.7 / 88.0 over the four tracks.

With the prior, the prior-aware models are the strongest single models, and the best system averages fourteen of them. It reaches 92.6 / 100 / 91.2 over the four tracks, above the development leader (91.4 / 100 / 89.7) on T1, T3, T4 and the average, and slightly below it on T2.

C. The negative class

Without the prior, the three reversible labels are rarely confused with each other; the errors that remain are almost all NEG against the rest. We tested five ways to improve that decision (Table II).

Class weighting (weight 2 on NEG) and focal loss ($\gamma = 2$) lowered HardCons on mDeBERTa by 3.9 and 1.1 points. Reverse-negative augmentation gained 0.8 F1 points on mDeBERTa but cost 4.7 SoftCons points, and hurt XLM-R-large on every metric. We then added 1,010 chunk pairs from the iSTS 2016 answers-students data, mapping EQUI to EQ, SPE1 and SPE2 to FE and BE, and SIMI, REL and OPPO to NEG. We did not use the headlines and images subsets, because they are the source of PhrasIS and contain its test pairs, and none of the converted pairs occurs in the DiCo training or development data. The extra data lowered F1 for both backbones, and a second mDeBERTa seed was also worse.

The last method was a stacked detector. A logistic regression and a gradient-boosted classifier predict NEG from the single-row log-probabilities of the seven ensemble members and from

TABLE III
DEVELOPMENT RESULTS WITHOUT THE STRUCTURAL PRIOR (F1 = WEIGHTED F1, S = SOFTCONS, H = HARDCONS, IN %). A = mDeBERTa, FINE-GRAINED VALIDATION; B = XLM-R-LARGE XNLI. SINGLE SEEDS EXCEPT ENSEMBLES. BEST VALUE PER COLUMN IN BOLD.

System	T1 EN			T2 ES			T3 EU			T4 Mixed			ALL-4		
	F1	S	H	F1	S	H	F1	S	H	F1	S	H	F1	S	H
mmBERT-base	77.4	87.7	78.3	74.1	91.7	78.0	67.8	88.8	73.6	72.5	83.4	74.4	72.9	87.9	76.1
mDeBERTa-v3-base NLI	83.6	95.3	89.2	79.3	94.2	86.6	77.4	93.5	84.1	78.9	94.2	85.9	79.8	94.3	86.5
mDeBERTa, fine-grained val. (A)	81.8	96.0	88.4	78.2	94.9	86.3	78.8	95.3	86.3	78.8	97.5	87.0	79.4	95.9	87.0
A + rev.-neg. aug.	83.4	92.1	86.3	81.1	92.4	84.5	76.4	89.2	79.1	80.0	91.3	82.7	80.2	91.2	83.1
A + NEG weight 2	82.8	92.4	85.9	80.5	93.9	85.9	76.7	88.4	79.4	78.7	90.3	81.2	79.7	91.2	83.1
A + focal ($\gamma=2$)	81.4	94.9	88.1	78.5	94.9	86.6	75.3	94.2	84.5	77.6	93.5	84.5	78.2	94.4	85.9
A + iSTS answers-students	80.6	96.0	88.4	76.8	95.3	85.6	76.2	95.7	85.9	77.5	97.8	88.1	77.8	96.2	87.0
A, symmetric arch.	82.1	93.1	85.6	78.6	93.5	84.1	77.2	92.4	82.7	79.4	94.2	84.8	79.3	93.3	84.3
XLM-R-large XNLI (B)	83.0	95.3	88.1	79.2	96.4	87.4	75.1	96.8	82.7	77.9	96.8	87.0	78.8	96.3	86.3
B + rev.-neg. aug.	78.5	92.1	83.0	75.1	93.9	80.9	70.1	91.0	75.5	73.3	93.9	79.1	74.3	92.7	79.6
B + iSTS answers-students	80.4	93.1	85.2	76.6	92.1	83.8	75.7	93.5	83.0	77.8	92.4	85.2	77.6	92.8	84.3
Ens. clean-7 (A, B + 5 seeds)	83.7	95.3	89.9	80.6	96.0	88.1	79.4	95.7	87.0	79.4	95.7	87.0	80.8	95.7	88.0

TABLE IV
DEVELOPMENT RESULTS WITH THE STRUCTURAL PRIOR, AND THE CODABENCH DEVELOPMENT LEADER ON 29 SEPTEMBER 2026. BEST VALUE PER COLUMN AMONG OUR SYSTEMS IN BOLD; COLUMNS WHERE ALL SYSTEMS TIE ARE NOT MARKED.

System	T1 EN			T2 ES			T3 EU			T4 Mixed			ALL-4		
	F1	S	H	F1	S	H	F1	S	H	F1	S	H	F1	S	H
mmBERT-base	87.1	100.0	84.8	85.2	100.0	82.7	82.1	100.0	79.8	88.5	100.0	86.6	85.7	100.0	83.5
mDeBERTa-v3-base NLI	92.3	100.0	91.0	90.5	100.0	88.8	89.6	100.0	87.7	90.9	100.0	89.2	90.8	100.0	89.2
mDeBERTa, fine-grained val. (A)	91.7	100.0	90.3	89.6	100.0	87.7	89.9	100.0	88.1	90.3	100.0	88.4	90.4	100.0	88.6
A + rev.-neg. aug.	92.6	100.0	91.3	90.9	100.0	89.2	87.0	100.0	84.5	90.9	100.0	89.2	90.4	100.0	88.5
A + NEG weight 2	92.1	100.0	90.6	90.6	100.0	88.8	87.9	100.0	85.6	90.0	100.0	88.1	90.2	100.0	88.3
A + focal ($\gamma=2$)	91.8	100.0	90.3	90.6	100.0	88.8	89.4	100.0	87.4	91.6	100.0	89.9	90.8	100.0	89.1
A + iSTS answers-students	92.1	100.0	90.6	89.6	100.0	87.7	89.4	100.0	87.4	90.6	100.0	88.8	90.4	100.0	88.6
A, symmetric arch.	91.2	100.0	89.5	89.6	100.0	87.7	88.2	100.0	85.9	90.7	100.0	88.8	89.9	100.0	88.0
XLM-R-large XNLI (B)	92.4	100.0	91.0	91.0	100.0	89.2	85.9	100.0	83.4	91.3	100.0	89.5	90.2	100.0	88.3
B + rev.-neg. aug.	91.0	100.0	89.2	88.3	100.0	85.9	85.3	100.0	82.3	86.6	100.0	83.8	87.8	100.0	85.3
B + iSTS answers-students	90.6	100.0	88.8	90.5	100.0	88.8	89.0	100.0	87.0	91.8	100.0	90.3	90.5	100.0	88.7
Ens. clean-7 (A, B + 5 seeds)	93.3	100.0	92.1	91.2	100.0	89.5	90.7	100.0	88.8	92.5	100.0	91.0	91.9	100.0	90.3
mDeBERTa, prior-aware	92.3	100.0	91.0	90.9	100.0	89.2	89.1	100.0	87.0	92.1	100.0	90.6	91.1	100.0	89.4
mDeBERTa, prior-aware, symmetric	93.0	100.0	91.7	91.8	100.0	90.3	90.1	100.0	88.1	92.5	100.0	91.0	91.9	100.0	90.3
XLM-R-large, prior-aware	91.8	100.0	90.3	91.0	100.0	89.2	88.5	100.0	86.3	91.9	100.0	90.3	90.8	100.0	89.0
XLM-R-large, prior-aware, symmetric	92.4	100.0	91.0	91.2	100.0	89.5	87.6	100.0	85.2	91.0	100.0	89.2	90.5	100.0	88.7
Ens. prior-aware-5	93.9	100.0	92.8	91.8	100.0	90.3	90.6	100.0	88.8	92.2	100.0	90.6	92.1	100.0	90.6
Ens. prior-aware-5 + clean-7	94.2	100.0	93.1	92.1	100.0	90.6	90.4	100.0	88.4	92.8	100.0	91.3	92.4	100.0	90.9
Ens. prior-14 (+ 2 symmetric)	94.2	100.0	93.1	92.4	100.0	91.0	91.3	100.0	89.5	92.5	100.0	91.0	92.6	100.0	91.2
Leaderboard #1	92.4	100	91.0	92.8	100	91.3	90.3	100	88.4	90.2	100	88.2	91.4	100	89.7

lexical features (token overlap, containment, length, number mismatch, negation words), with five-fold cross-validation grouped by `pair_id`. The best setting raised F1 by 0.6 points but lowered SoftCons by 3.9 and HardCons by 3.3. The ensemble’s own NEG score seems to carry the same lexical signal already.

Moving the NEG threshold β only trades one score for another (Fig. 3). Lowering it towards -1.5 lifts HardCons to 90.3 and SoftCons to 99.3 while F1 falls to 76.9; raising it does the reverse. The mean of the three scores stays between 88 and 89 over the whole range.

D. Cross-track pooling

Because a `pair_id` is shared by all tracks, decoding the four development files together pools up to eighteen views of each source pair. In a four-model ensemble with the prior this added about one point of both F1 and HardCons over per-

track decoding (92.7 / 100 / 91.3 against 91.8 / 100 / 90.3). It also uses the other tracks’ inputs to predict a given track and makes the four tracks’ predictions identical, so we leave it out of the reported results until the organizers say whether it is allowed.

VII. DISCUSSION

Two caveats apply to every number above. First, all model, ensemble and threshold choices were made on the development set, which has 277 reversible pairs per track; the standard error of HardCons at that size is about 2.6 points, so differences of one or two points between single runs are within noise, and the reported scores are optimistic. Grouped cross-validation over the training and development data would give firmer estimates. Second, the lead over the leaderboard depends entirely on the structural prior. Without it, our best

system is about eleven points of F1 behind, and the gap is the NEG decision.

The negative results are consistent with what is known about NLI models on short, lexically overlapping inputs [11], [12]: the hard NEG cases are phrases that are related, similar or opposite, and more weight, more data of the same kind or extra surface features did not teach the models to separate them. Methods that change the model’s inductive bias rather than its training signal, such as larger instruction-tuned models used as teachers, are the obvious next step.

VIII. CONCLUSION

Consistency in DiCo-NLI can be guaranteed by decoding the two directions of a pair together, which raises SoftCons and HardCons at no training cost. With that in place, the task reduces to deciding whether a phrase pair stands in an entailment relation at all. Our clean seven-model ensemble reaches 80.8 / 95.7 / 88.0 on the development set; the same models with a layout-based prior reach 92.6 / 100 / 91.2. Whether that prior may be used in the evaluation phase is a question for the organizers.

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